SENTINELT
Robo Wars RC Combat Robot
Holistic Build Document

Designed and developed by Thato Glen Assegaai

Independent project ownership and engineering development

Verified CAD envelope: 318.16 mm x 495.00 mm x 255.41 mm | Footprint audit: 180/180 frames PASS

10. Bill of Materials and Cost

The Bill of Materials records component name, quantity, unit cost, supplier, stock code, direct purchase link and total line cost. The current priced component subtotal is **R8,039.15**. The guarded active-mechanism motor/controller remains intentionally marked TBD until its electrical and mass compatibility is finalized.

Core item	Cost status
6 mm HDPE chassis/armor stock	Priced
Drive motors and wheels	Priced
2.4 GHz transmitter + receiver	Priced
3S LiPo battery	Priced
Dual brushed drive ESC	Priced
Main isolation, E-Stop, relay, fuse and distribution	Priced
High-current wiring and fasteners	Priced
Guarded active-mechanism motor/controller	TBD
Current priced component subtotal	R8,039.15

11. Buildability and Serviceability

SentinelT is designed as a modular build. The top and service panels provide access to the battery, receiver, ESC, power distribution, main isolation and E-Stop control hardware. The selected direct-drive wheels mount to 6 mm motor shafts, reducing drivetrain part count.

Figure 5. Side CAD view showing the compact chassis envelope.

12. Current Development Status

The digital robot design, external CAD geometry, dimensional simulation, source-code framework, safety architecture and procurement plan are documented. The next CAD revision will add simplified internal component models, wiring paths, the 2.4 GHz transmitter model, a cutaway view and an exploded internal-layout view.

Physical claims remain deliberately separated from digital results. Final physical mass, current draw, thermal behaviour, E-Stop interruption performance and drivetrain response will be measured only after fabrication and bench testing.

13. Project Ownership and Responsibilities

SentinelT is independently designed and developed by **Thato Glen Assegaai**. My responsibilities include Robo Wars system concept and architecture, Blender CAD and mechanical packaging, Python footprint and mass-budget tools, 2.4 GHz RC control framework, power and E-Stop safety architecture, component research, Bill of Materials, procurement planning, and technical documentation.

Third-party commercial components are identified by manufacturer and supplier in the Bill of Materials. Their inclusion does not imply ownership of those products; the SentinelT system design, integration plan, CAD work, control framework and documentation are presented as my project work.